

GLEICHEN AGCM, AB, Canada

WMO: 714980

Lat: 50.9345N

Lon: 112.9408W

Elev: 2976

StdP: 13.18

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.6	-15.3	-26.4	1.4	-20.5	-20.7	2.0	-15.0	27.9	27.8	25.1	27.0	6.5	210	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.4	85.1	61.5	81.9	60.3	78.7	59.6	65.6	77.9	63.8	76.4	62.1	74.5	8.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.3	90.5	71.4	59.0	83.3	69.8	57.0	77.5	67.3	32.1	78.7	30.7	76.4	29.3	74.6	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	21.2	18.7	-26.8	91.4	4.2	3.6	-29.9	94.0	-32.3	96.1	-34.7	98.1	-37.8	100.7		
(5)			-27.0	70.6	4.2	2.3	-30.0	72.3	-32.5	73.6	-34.8	74.9	-37.8	76.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.2	17.6	14.8	26.4	39.5	51.2	58.2	62.9	62.0	53.8	40.8	26.7	14.8
	DBStd	20.82	14.63	15.11	14.67	8.97	7.91	5.51	4.76	5.69	8.59	9.47	13.33	14.52
	HDD50	5301	1004	986	733	329	83	6	0	2	57	308	701	1092
	HDD65	9503	1469	1406	1197	766	431	213	97	125	344	750	1148	1557
	CDD50	1354	0	0	0	12	119	252	400	373	172	23	3	0
	CDD65	82	0	0	0	0	2	9	31	31	9	0	0	0
	CDH74	1958	0	0	0	14	140	210	617	676	291	10	0	0
	CDH80	481	0	0	0	1	19	37	152	196	76	0	0	0
	WSAvg	8.4	9.5	8.5	9.2	9.4	9.6	8.8	6.5	6.1	7.6	8.3	8.6	8.5
	PrecAvg	14.3	0.6	0.4	0.6	1.0	1.9	2.5	2.0	1.6	1.7	0.5	0.6	0.6
(15) Precipitation	PrecMax	20.1	1.9	1.0	1.6	3.0	4.3	6.4	6.0	3.7	6.2	1.5	2.9	1.5
	PrecMin	9.6	0.0	0.0	0.1	0.0	0.3	0.3	0.6	0.2	0.1	0.0	0.0	0.0
	PrecStd	3.0	0.4	0.2	0.4	0.9	1.0	1.5	1.3	0.9	1.4	0.4	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	48.9	61.3	73.9	82.8	84.3	88.6	89.7	86.9	74.4	64.1	49.1
		MCWB	38.3	39.8	45.9	51.5	55.2	60.9	65.7	61.0	58.6	53.0	47.7	39.4
	2%	DB	42.6	42.5	55.9	67.2	78.2	79.5	84.2	85.2	82.0	66.6	54.7	42.3
		MCWB	36.5	36.1	42.6	47.9	54.2	59.1	64.1	61.9	56.9	50.0	42.6	36.2
	5%	DB	39.9	38.2	50.4	61.1	73.6	75.7	80.6	81.4	76.9	61.8	48.6	38.4
		MCWB	35.2	33.4	39.9	44.1	51.9	57.6	63.2	61.0	55.7	47.3	39.9	33.5
	10%	DB	37.2	34.6	44.4	56.0	68.9	72.4	77.1	77.5	71.5	56.6	43.7	34.6
		MCWB	33.0	31.3	37.1	41.7	50.3	56.1	62.1	59.9	53.9	44.8	36.2	30.5
	0.4%	WB	39.5	40.8	46.9	51.8	56.7	64.9	70.1	66.5	60.7	54.6	47.7	40.3
		MCDB	45.6	46.9	58.9	72.1	77.0	77.1	82.0	79.2	80.2	71.7	62.2	46.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.9	36.4	43.6	48.3	55.0	61.9	66.8	64.6	58.4	51.1	43.6	36.5
		MCDB	41.7	41.8	54.3	64.8	73.9	74.7	78.7	79.0	76.8	63.9	53.2	42.4
	5%	WB	35.4	33.9	40.3	45.2	53.1	59.6	64.8	63.0	56.3	48.2	40.4	33.7
		MCDB	39.7	37.8	49.3	58.8	70.5	71.3	76.4	77.1	73.8	60.2	47.9	38.5
	10%	WB	33.2	31.3	37.4	42.7	51.3	57.7	63.2	61.1	54.5	45.6	37.0	30.8
		MCDB	37.2	34.4	44.0	54.1	65.7	69.3	74.7	74.0	70.3	55.6	42.4	34.3
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	17.7	19.5	22.8	24.7	23.0	25.4	26.2	25.9	23.0	18.8	16.6
		MCDBR	16.9	20.9	27.1	33.2	33.1	29.4	32.2	34.2	35.8	31.0	25.8	19.6
		MCWBR	12.6	15.6	15.8	16.5	13.5	12.8	15.7	14.4	14.6	15.9	15.5	14.7
	5% WB	MCDBR	16.9	20.0	25.4	31.1	28.9	25.7	28.4	29.4	32.6	29.7	25.0	20.1
		MCWBR	12.8	15.2	15.8	16.2	12.4	12.9	15.2	13.9	13.9	15.5	15.7	15.4
(40) Clear-Sky Solar Irradiance	taub		0.260	0.266	0.285	0.338	0.359	0.351	0.351	0.368	0.323	0.282	0.254	0.238
	taud		2.389	2.451	2.468	2.323	2.318	2.387	2.408	2.344	2.441	2.533	2.490	2.414
	Ebn at Noon		250	281	295	286	284	287	285	273	276	272	254	244
	Edh at Noon		19	23	28	36	38	36	35	35	29	21	17	16
(44) All-Sky Solar Radiation	RadAvg		370	688	1114	1520	1825	1980	2137	1717	1213	741	401	282
	RadStd		28	62	88	111	125	122	105	115	118	85	35	26

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (82 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page